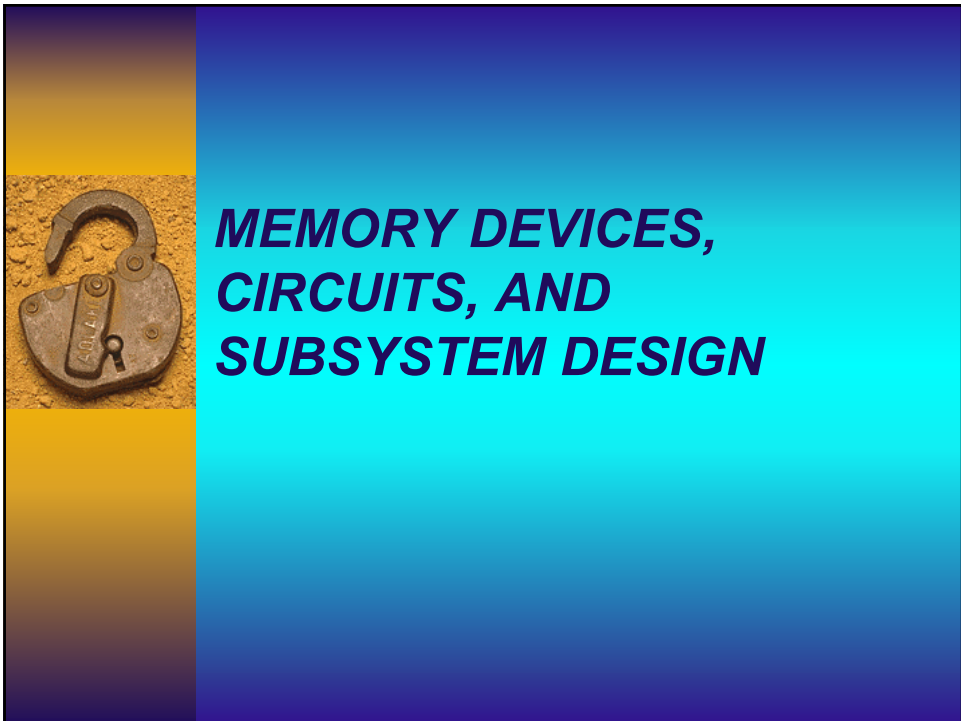




MEMORY DEVICES, CIRCUITS, AND SUBSYSTEM DESIGN

A vertical yellow bar on the left side of the slide, with a key resting on a textured, brownish surface.

MEMORY DEVICES, CIRCUITS, AND SUBSYSTEM DESIGN

- 9.1 Program and Data Storage
- 9.2 Read-Only Memory
- 9.3 Random Access Read/Write Memories
- 9.4 Parity, the Parity Bit, and Parity-Checker/Generator Circuit
- 9.5 FLASH Memory
- 9.6 Wait-State Circuitry
- 9.7 8088/8086 Microcomputer System Memory Circuitry

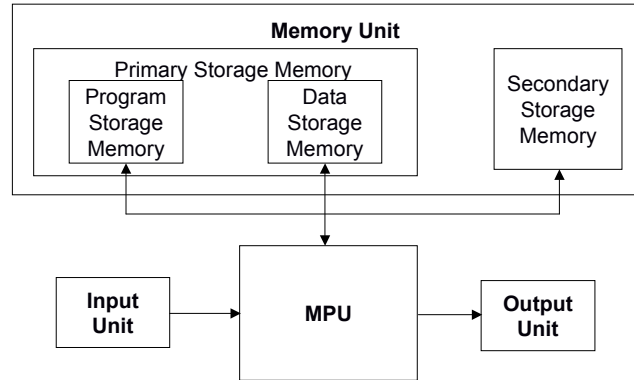
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The logo of National Taiwan University, featuring a stylized 'N' and 'T' inside a hexagon, with a gear and a lightning bolt.

9.1 Program and Data Storage

- The memory unit of a microcomputer is partitioned into a **primary storage section** and **secondary storage section**.



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9.1 Program and Data Storage

- The **basic input/output system (BIOS)** are programs held in ROM. They are called **firmware** because of their permanent nature.
- The typical size of a BIOS ROM used in a PC today is 256 Kbytes.
- Programs are normally read in from the secondary memory storage device, stored in the program storage part of memory, and then run.

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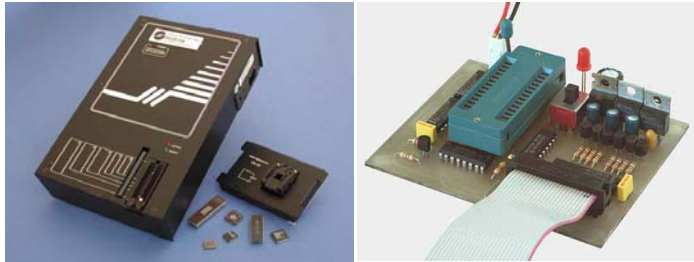




9.2 Read-Only Memory

■ ROM, PROM, and EPROM

- ❖ Mask-programmable read-only memory (ROM)
- ❖ One-time-programmable read-only memory (PROM)
- ❖ Erasable read-only memory (EPROM)



EPROM Programming unit

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9.2 Read-Only Memory

■ ROM, PROM, and EPROM



EPROM erasing unit

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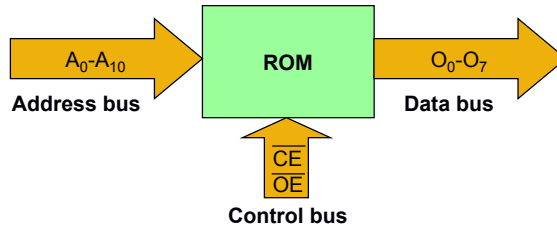
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9.2 Read-Only Memory

■ Block diagram of a read-only memory

- ❖ Address bus
- ❖ Data bus
- ❖ Control bus
 - Chip enable ($\overline{CE}$)
 - Output enable ($\overline{OE}$)



Block diagram of a ROM

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9.2 Read-Only Memory

EXAMPLE

Suppose the block diagram in the previous slide had 15 address lines and eight data lines. How many bytes of information can be stored in the ROM? What is its total storage capacity?

Solution:

With 8 data lines, the number of bytes is equal to the number of locations, which is

$$2^{15} = 32,768 \text{ bytes}$$

This gives a total storage of

$$32,768 \times 8 = 262,144 \text{ bits}$$

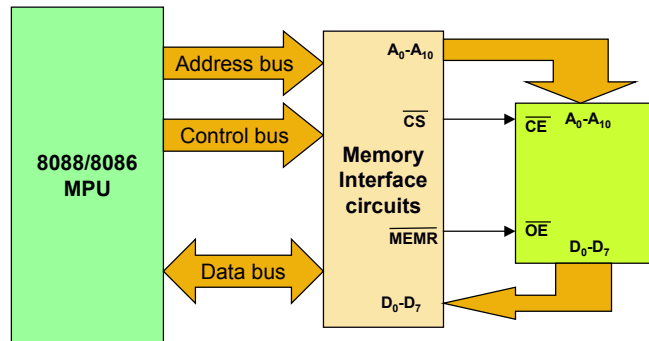
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9.2 Read-Only Memory

■ Read operation



Read-only memory interface



9.2 Read-Only Memory

■ Standard EPROM ICs

EPROM	Density (bits)	Capacity (bytes)
2716	16K	2Kx8
2732	32K	4Kx8
27C64	64K	8Kx8
27C128	128K	16Kx8
27C256	256K	32Kx8
27C512	512K	64Kx8
27C010	1M	128Kx8
27C020	2M	256Kx8
27C040	4M	512Kx8

Standard EPROM devices



A close-up photograph of a single, dark, ornate skeleton key. The key features a prominent, looped bow at the top and a long, slender shaft that tapers slightly towards the bit. The bit itself is simple and notched. The key is set against a textured, light-colored background, possibly a piece of stone or concrete.

- Standard EPROM ICs

V _{PP}	1	28	V _{CC}
A ₁₂	2	27	A ₁₄
A ₇	3	26	A ₁
A ₆	4	25	A ₈
A ₅	5	24	A ₉
A ₄	6	23	A ₁
A ₃	7	22	$\overline{\text{OE}}$
A ₂	8	21	A ₁₅
A ₁	9	20	$\overline{\text{CE}}$
A ₀	10	19	O ₇
O ₀	11	18	O ₆
O ₁	12	17	O ₅
O ₂	13	16	O ₄
Gnd	14	15	O ₃

Pin layouts of standard EPROMs.

A large, ornate metal key with a circular bow and a notched bit, resting on a textured, yellowish-brown surface.

- Standard EPROM ICs

-

9.2 Read-Only Memory

■ Standard EPROM ICs

Versions		$V_{CC} \pm 5\%$		27C256-120V05		27C256-135V05		27C256-150V05		27C256-1 P27C256-1 N27C256-1		27C256-2 P27C256-2 N27C256-2		27C256 P27C256 N27C256		Unit
		$V_{CC} \pm 10\%$				27C256-135V10		27C256-150V10				27C256-20 P27C256-20 N27C256-20		27C256-25 P27C256-25 N27C256-25		
Symbol	Parameter	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
t_{ACC}	Address to output delay		120		135		150		170		200		250			ns
t_{OE}	$\overline{CE}$ to output delay		120		135		150		170		200		250			ns
t_{OE}	$\overline{OE}$ to output delay		60		65		70		75		75		100			ns
$t_{OE}^{(2)}$	$\overline{OE}$ high to output high-Z		30		35		45		55		55		60			ns
$t_{OH}^{(2)}$	Output hold from addresses, $\overline{CE}$ or $\overline{OE}$ change—whichever is first	0		0		0		0		0		0		0		ns

Notes:

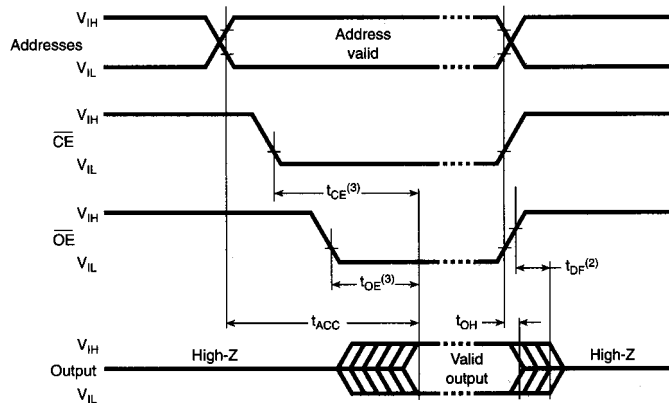
1. A.C. characteristics tested at $V_{IH} = 2.4$ V and $V_{IL} = 0.45$ V.
Timing measurements made at $V_{OL} = 0.8$ V and $V_{OH} = 2.0$ V.
2. Guaranteed and sampled.
3. Package Prefixes: No Prefix = CERDIP; N = PLCC; P = Plastic DIP.

EPROM device timing characteristics



9.2 Read-Only Memory

■ Standard EPROM ICs



EPROM switching waveforms





9.2 Read-Only Memory

■ Standard EPROM ICs

- ❖ A complex series of program and verify operations are performed to program each storage location in an EPROM.
- ❖ The two widely used programming sequences are the Quick-Pulse Programming Algorithm and the Intelligent Programming Algorithm.
- ❖ CMOS EPROMs are designed to provide TTL-compatible input and output logic level.

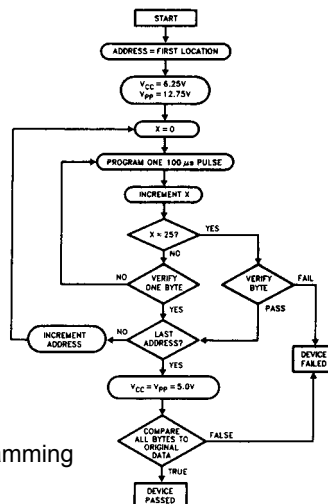
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9.2 Read-Only Memory

■ Standard EPROM ICs



Quick-Pulse Programming
Algorithm flowchart

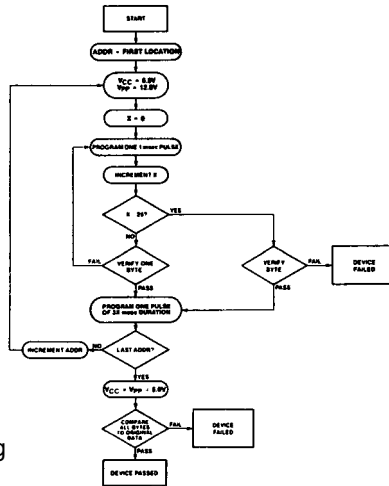
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9.2 Read-Only Memory

■ Standard EPROM ICs



Intelligent Programming
Algorithm flowchart

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9.2 Read-Only Memory

■ Standard EPROM ICs

Symbol	Parameter	Notes	Min	Typ ⁽¹⁾	Max	Unit	Test Conditions
I_{L1}	Input load current			0.01	1.0	μA	$V_{IH} = 0V$ to V_{CC}
I_{LO}	Output leakage current				± 10	μA	$V_{OH} = 0V$ to V_{CC}
I_{PP}	V_{PP} read current	5			200	μA	$V_{PP} = V_{CC}$
I_{SR1}	V_{CC} current standby	TTL 8			1.0	mA	$CE = V_{IH}$
I_{SR2}		CMOS 4			100	μA	$CE = V_{CC}$
I_{OC1}	V_{CC} current active	5, 8			30	mA	$CE = V_{L1}$ $f = 5 \text{ MHz}$
V_L	Input low voltage ($\pm 10\%$ supply) (TTL)		-0.5		0.8	V	
	Input low voltage (CMOS)		-0.2		0.8		
V_{IH}	Input high voltage ($\pm 10\%$ supply) (TTL)		2.0		$V_{CC} + 0.5$	V	
	Input high voltage (CMOS)		$0.7 V_{CC}$		$V_{CC} + 0.2$		
V_{OL}	Output low voltage				0.45	V	$I_{OL} = 2.1 \text{ mA}$
V_{OH}	Output high voltage		3.5			V	$I_{OH} = -2.5 \text{ mA}$
I_{OS}	Output short circuit current	6			100	mA	
V_{PP}	V_{PP} read voltage	7	$V_{CC} - 0.7$		V_{CC}	V	

Notes:

1. Minimum D.C. input voltage is -0.5V. During transitions, the inputs may undershoot to -2.0V for periods less than 20 ns. Maximum D.C. voltage on output pins is $V_{CC} + 0.5V$ which may overshoot to $V_{CC} + 2V$ for periods less than 20 ns.
2. Operating temperature is for commercial product defined by this specification. Extended temperature options are available in: EXPRESS and Military version.
3. Typical limits are at $V_{CC} = 5V$, $T_A = +25^\circ C$.
4. CE is $V_{CC} \pm 0.2V$. All other inputs can have any value within spec.

5. Maximum Active power usage is the sum $I_{PP} + I_{OC}$. The maximum current value is with outputs Q_0 to Q_7 unloaded.
6. Output shorted for no more than one second. No more than one output shorted at a time. I_{OS} is sampled but not 100% tested.
7. V_{PP} may be one diode voltage drop below V_{CC} . It may be connected directly to V_{CC} . Also, V_{PP} must be applied simultaneously or before V_{PP} and removed simultaneously or after V_{PP} .
8. V_{L1} , V_{H1} levels at TTL inputs.

DC electrical characteristics of the 27C256

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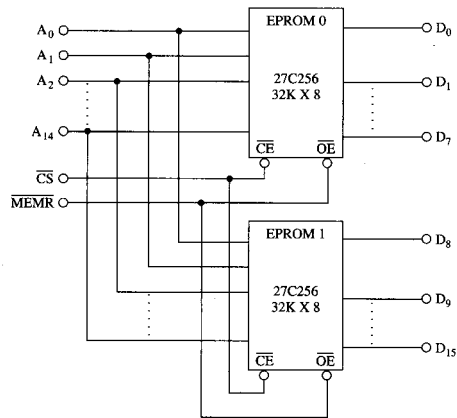
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9.2 Read-Only Memory

- Expanding EPROM word length and word capacity

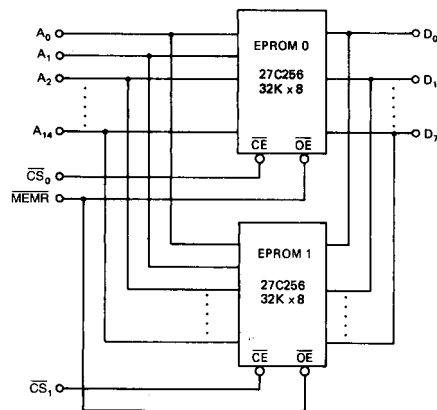


Expanding word length



9.2 Read-Only Memory

- Expanding EPROM word length and word capacity



Expanding word capacity



9.3 Random Access Read/Write Memories

- The memory section of a microcomputer system is normally formed from both read-only memories and random access read/write memories (RAM)
- RAM is different from ROM in two ways:
 - ❖ Data stored in RAM is not permanent in nature.
 - ❖ RAM is volatile – that is, if power is removed from RAM, the stored data are lost.
- RAM is normally used to store data and application programs for execution.



9.3 Random Access Read/Write Memories

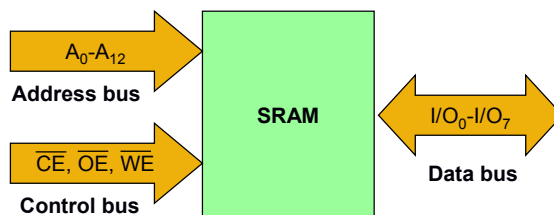
- Static and dynamic RAMs
 - ❖ For a static RAM (SRAM), data remain valid as long as the power supply is not turned off.
 - ❖ For a dynamic RAM (DRAM), we must both keep the power supply turned on and periodically restore the data in each location.
 - ❖ The recharging process is known as **refreshing** the DRAM.



9.3 Random Access Read/Write Memories

■ Block diagram of a static RAM

- ❖ The most commonly used densities in RAM IC system designs are the 64KB and 256KB devices.
- ❖ The data lines are bidirectional and the read/write operations are controlled by the $\overline{CE}$, $\overline{OE}$, $\overline{WE}$ control signals.

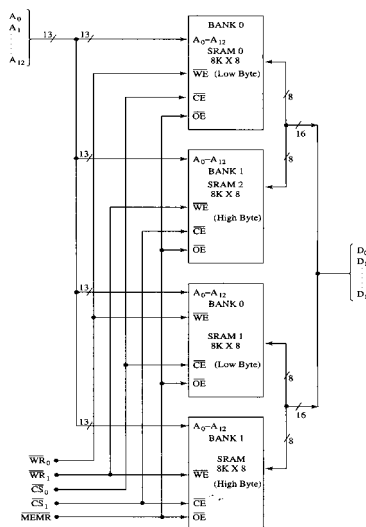


Block diagram of a static RAM



9.3 Random Access Read/Write Memories

■ A static RAM system



16K x 16-bit SRAM circuit



9.3 Random Access Read/Write Memories

■ Standard static RAM ICs

SRAM	Density (bits)	Organization
4361	64K	64Kx1
4363	64K	16Kx4
4364	64K	8Kx8
43254	256K	64Kx4
43256A	256K	32Kx8
431000A	1M	128Kx8

Part number	Read/write cycle time
4364-10	100 ns
4364-12	120 ns
4364-15	150 ns
4364-20	200 ns

Speed selection for the 4364 SRAM

Standard SRAM devices

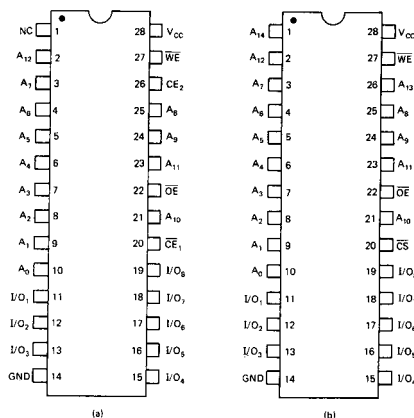
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9.3 Random Access Read/Write Memories

■ Standard static RAM ICs



(a) 4365 pin layout. (b) 43256A pin layout

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9.3 Random Access Read/Write Memories

Parameter	Symbol	Limits			Unit	Test Conditions
		Min	Typ	Max		
Input leakage current	I_{LI}			1	μA	$V_{IN} = 0 \text{ V to } V_{CC}$
Output leakage current	I_{LO}			1	μA	$V_{IO} = 0 \text{ V to } V_{CC}$ $CE_1 = V_{IH}$ or $CE_2 = V_{IL}$ or $OE = V_{IH}$ or $WE = V_{IL}$
Operating supply current	I_{CCA1}			(1)	mA	$CE_1 = V_{IL}$, $CE_2 = V_{IH}$, $I_{IO} = 0$, Min cycle
	I_{CCA2}		5	10	mA	$CE_1 = V_{IL}$, $CE_2 = V_{IH}$, $I_{IO} = 0$, DC current
	I_{CCA3}		3	5	mA	$CE_1 \leq 0.2 \text{ V}$, $CE_2 \geq V_{CC} - 0.2 \text{ V}$, $V_{IL} \leq 0.2 \text{ V}$, $V_{IH} \geq V_{CC} - 0.2 \text{ V}$, $f = 1 \text{ MHz}$, $I_{IO} = 0$
Standby supply current	I_{SB}			(2)	mA	$CE_1 \geq V_{IH}$ or $CE_2 = V_{IL}$
	I_{SB1}			(3)	mA	$CE_1 \geq V_{CC} - 0.2 \text{ V}$ $CE_2 \geq V_{CC} - 0.2 \text{ V}$
	I_{SB2}			(3)	mA	$CE_2 \leq 0.2 \text{ V}$
Output voltage, low	V_{OL}			0.4	V	$I_{OL} = 2.1 \text{ mA}$
Output voltage, high	V_{OH}	2.4			V	$I_{OH} = -1.0 \text{ mA}$

Notes:

- (1) $\mu PD4364-10/10L$: 45 mA max
 $\mu PD4364-12/12L/12LL$: 40 mA max
 $\mu PD4364-15/15L/15LL$: 40 mA max
 $\mu PD4364-20/20L/20LL$: 35 mA max
 (2) $\mu PD4364-xx$: 5 mA max
 $\mu PD4364-xxL$: 3 mA max
 (3) $\mu PD4364-xx$: 2 mA max
 $\mu PD4364-xxL$: 100 μA max
 $\mu PD4364-xxLL$: 60 μA max

DC electrical characteristics of the 4364

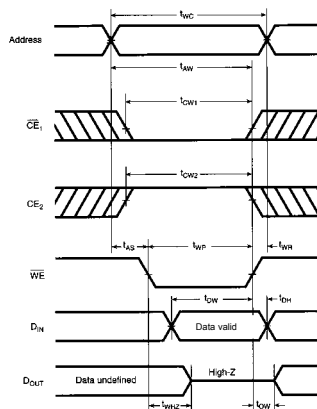
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9.3 Random Access Read/Write Memories

SRAM read and write cycle operation



Notes:

1. A write occurs during the overlap of a low CE_1 and a high CE_2 and a low WE .
2. CE_1 or WE (or CE_2) must be high (low) during any address transaction.
3. If OE is high the I/O pins remain in a high-impedance state.

Write-cycle timing diagram

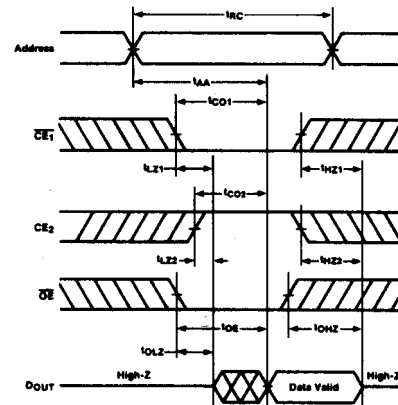
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9.3 Random Access Read/Write Memories

■ SRAM read and write cycle operation



Read-cycle timing diagram

Notes:
 [1] WE is a high for a read cycle.
 [2] The address inputs are valid prior to or coincident with the CE₁ transition low and the CE₂ transition high.

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9.3 Random Access Read/Write Memories

■ Standard dynamic RAM ICs

- ❖ Dynamic RAMs are available in higher densities than static RAMs. The most widely used DRAMs are the 64K-bit, 256K-bit, 1M-bit, and 4M-bit devices.
- ❖ Benefits of using DRAMs over SRAMs are:
 - Cost less
 - Consume less power
 - The 16- and 18-pin package take up less space
- ❖ To maintain the data in a DRAM, each of the rows of the storage array must typically be refreshed periodically, such as every 2 ms.

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9.3 Random Access Read/Write Memories

■ Standard dynamic RAM ICs

SRAM	Density (bits)	Organization
2164B	64K	64Kx1
21256	256K	256Kx1
21464	256K	64Kx4
421000	1M	1Mx4
424256	1M	256Kx4
44100	4M	4Mx1
44400	4M	1Mx4
44160	4M	256Kx16
416800	16M	8Mx2
416400	16M	4Mx4
416160	16M	1Mx16

Standard DRAM devices

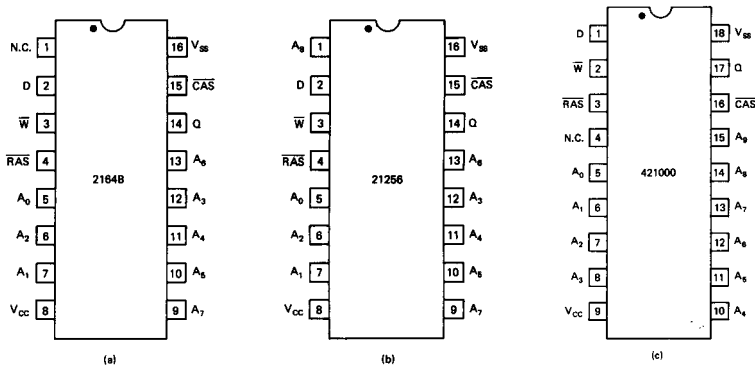
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9.3 Random Access Read/Write Memories

■ Standard dynamic RAM ICs



(a) 2164B pin layout. (b) 21256 pin layout. (c) 421000 pin layout

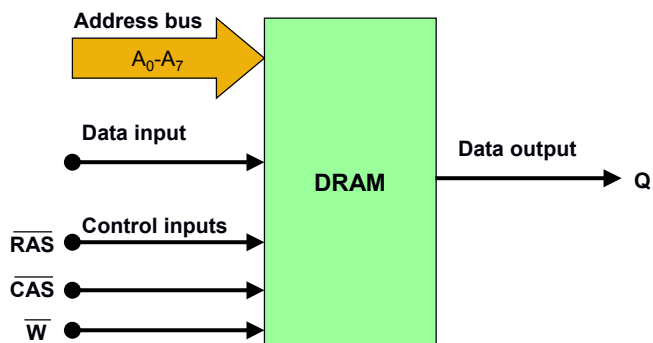
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9.3 Random Access Read/Write Memories

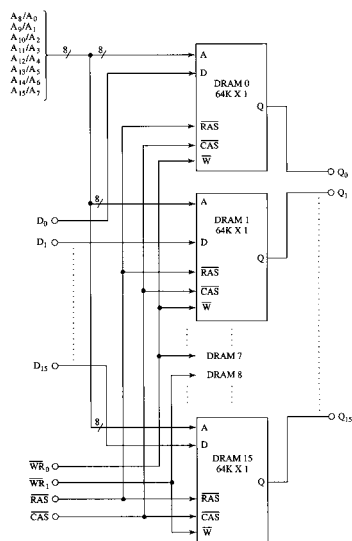
■ Standard dynamic RAM ICs



Block diagram of the 2164 DRAM



9.3 Random Access Read/Write Memories



64K x 16-bit DRAM circuit



9.3 Random Access Read/Write Memories

■ Evolution of RAM

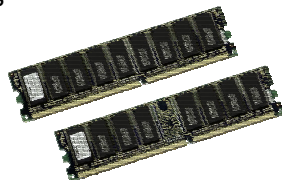
1970	RAM / DRAM	4.77 MHz
1987	FPM	20 MHz
1995	EDO	20 MHz
1997	PC66 SDRAM	66 MHz
1998	PC100 SDRAM	100 MHz
1999	RDRAM	800 MHz
1999/2000	PC133 SDRAM	133 MHz
2000	DDR SDRAM	266 MHz
2001	EDRAM	450MHz



9.3 Random Access Read/Write Memories

■ Evolution of RAM

- ❖ FPM-Fast Page Mode DRAM
 - traditional DRAM
- ❖ EDO-Extended Data Output
 - increases the Read cycle between Memory and the CPU
- ❖ SDRAM-Synchronous DRAM
 - synchronizes itself with the CPU bus and runs at higher clock speeds



9.3 Random Access Read/Write Memories

■ Evolution of RAM

❖ RDRAM-Rambus DRAM

-DRAM with a very high bandwidth (1.6 GBps)

❖ EDRAM-Enhanced DRAM

-(dynamic or power-refreshed RAM) that includes a small amount of static RAM (SRAM) inside a larger amount of DRAM so that many memory accesses will be to the faster SRAM. EDRAM is sometimes used as L1 and L2 memory and, together with Enhanced Synchronous Dynamic DRAM, is known as cached DRAM.

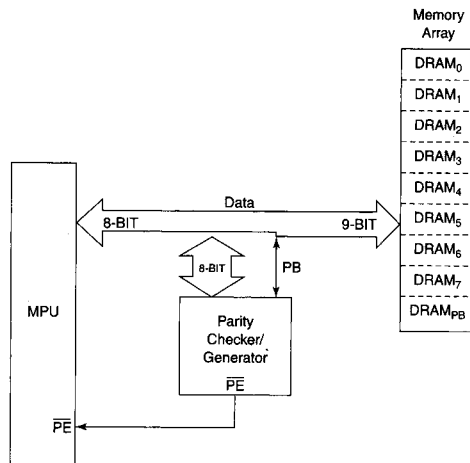


9.4 Parity, the Parity Bit, and Parity-Checker/Generator Circuit

- To improve the reliability of information transfer between the MPU and memory, a parity bit can be added to each byte of data.
- The **parity-checker/generator circuit** can be set up to produce either **even parity** or **odd parity**.
- The parity-check/generator signals parity error to MPU by setting PE to zero.
- In a 16-bit microcomputer system, there are normally two 8-bit banks of DRAM ICs in the data-storage memory array. A parity bit DRAM is added to each bank.



9.4 Parity, the Parity Bit, and Parity-Checker/Generator Circuit



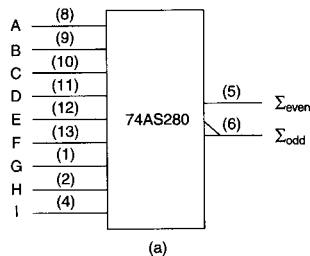
Data-storage memory interface with parity-checker generator

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9.4 Parity, the Parity Bit, and Parity-Checker/Generator Circuit



NUMBER OF INPUTS A THRU I THAT ARE HIGH	OUTPUTS	
	Σ EVEN	Σ ODD
0,2,4,6,8	H	L
1,3,5,7,9	L	H

(b)

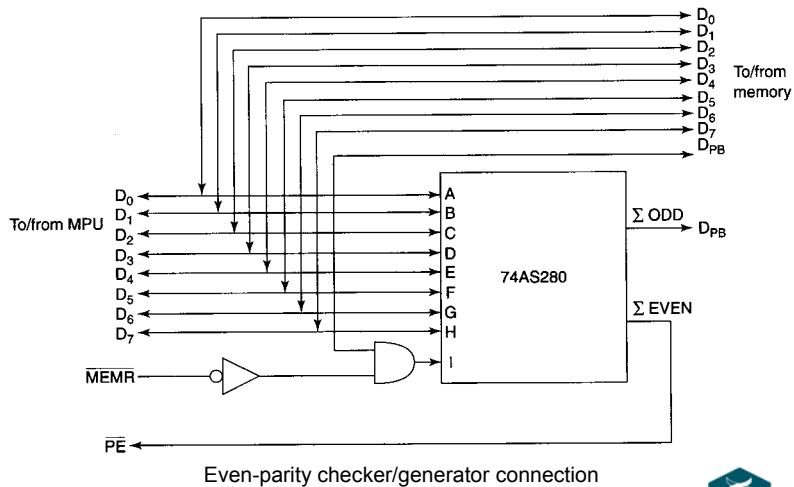
(a) Block diagram of the 74AS280. (b) Function table.

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9.4 Parity, the Parity Bit, and Parity-Checker/Generator Circuit



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9.5 FLASH Memory

- Flash memory devices are similar to EPROMs in that they are nonvolatile, are read like an EPROM, and program with an EPROM-like algorithm.
- The key difference between a FLASH memory and an EPROM is that its memory cells are erased electrically, instead of by exposure to ultraviolet light.
- When an erase operation is performed on a FLASH memory, either the complete memory array or a large block of storage location, not just one byte, is erased.
- The erase process of FLASH memory is complex and can take as long as several seconds.
- The FLASH memories find their widest use in microcomputer systems for storage of firmware.

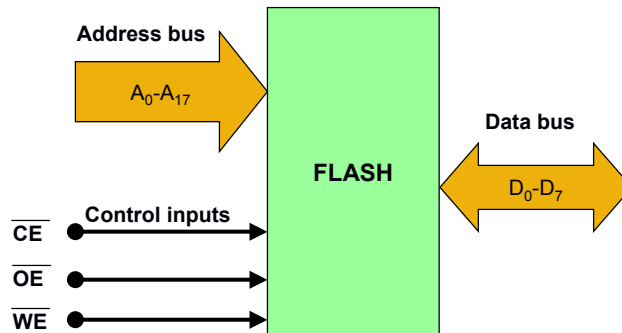
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9.5 FLASH Memory

■ Block diagram of a FLASH memory

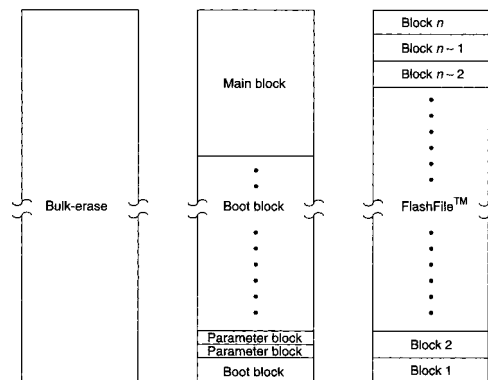


Block diagram of a FLASH memory



9.5 FLASH Memory

■ Bulk-erase, boot block, and FlashFile FLASH memory



FLASH memory array architectures



9.5 FLASH Memory

■ Standard bulk-erase FLASH memories

FLASH	Density (bits)	Capacity (bytes)
28F256	256K	32Kx8
28F512	512K	64Kx8
28F010	1M	128Kx8
28F020	2M	256Kx8

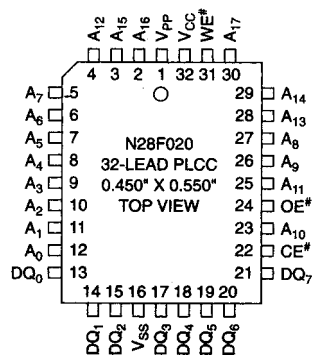
Standard bulk-erase FLASH memory devices



9.5 FLASH Memory

■ Standard bulk-erase FLASH memories

- ◆ The most popular package for housing FLASH memory ICs is the plastic leaded chip carrier, or PLCC.



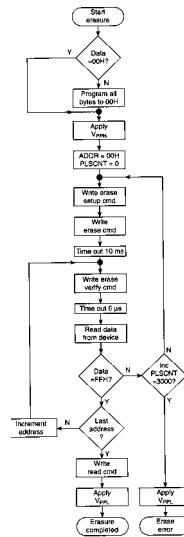
Pin layout of the 28F020.

Part number	Access time
28F020-70	70 ns
28F020-90	90 ns
28F020-120	120 ns
28F020-150	150 ns

Standard speed selection for the 28F020



9.5 FLASH Memory



Quick-erase algorithm of the 28F020.

Bus operation	Command	Comments
		Entire memory must = 00H before erase
		Use quick-pulse programming algorithm
Standby		Wait for V_{pp} ramp to V_{Vpp}
Write	Setup erase	Data = 20H
Write	Erase	Data = 20H
Standby		Duration of erase operation (user)
Write	Erase verify	Addr = byte to verify; Data = 00H; stops erase operation
Standby		Read byte to verify
Read		Read byte to verify
Standby		Compare output to FFH increment pulse-count
Write	Read	Data = 00H; resets the register for read operations
Standby		Wait for V_{pp} ramp to V_{Vpp}



9.5 FLASH Memory

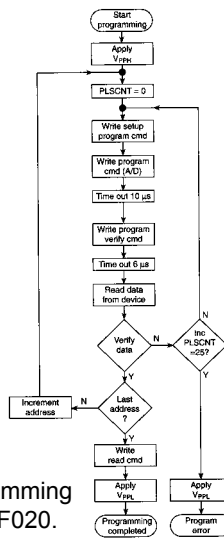
Standard bulk-erase FLASH memories

Command	Bus Cycles Req'd	First Bus Cycle			Second Bus Cycle		
		Operation	Address	Data	Operation	Address	Data
Read memory	1	Write	X	00H			
Read intelligent identifier codes	3	Write	X	90H	Read		
Setup erase/erase	2	Write	X	20H	Write	X	20H
Erase verify	2	Write	EA	A0H	Read	X	EVD
Setup program/program	2	Write	X	40H	Write	PA	PD
Program verify	2	Write	X	C0H	Read	X	PVD
Reset	2	Write	X	FFH	Write	X	FFH

28F020 command definitions



9.5 FLASH Memory



Bus operation	Command	Comments
Standby		Wait for V_{pp} ramp to V_{ppH}
		Initialize pulse-count
Write	Setup program	Data = 40H
Write	Program	Valid address/data
Standby		Duration of program operation (t_{pwr})
Write	Program verify	Data = C0H; stops program operation
Standby		t_{wex}
Read		Read byte to verify programming
Standby		Compare data output to data expected
Write	Read	Data = 00H; resets the register for read operations
Standby		Wait for V_{pp} ramp to V_{ppL}

Quick-pulse programming algorithm of the 28F020.

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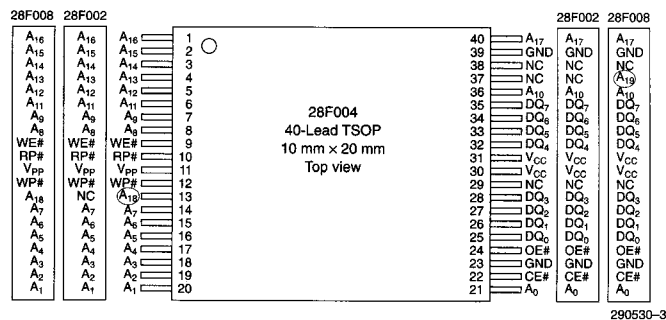
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9.5 FLASH Memory

■ Standard boot block FLASH memories

- ❖ The boot block FLASH memories are designed for used in embedded microprocessor application.



Pin-layout comparison of the TSOP 28F002, 28F004, and 28F008 IC

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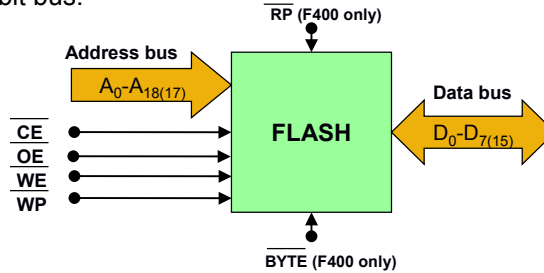
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9.5 FLASH Memory

■ Standard boot block FLASH memories

- ❖ One of the important features of boot block FLASH memory is what is known as **SmartVoltage**. This capability enables the device to be programmed with either a 5-V or 12-V value of V_{pp} .
- ❖ The boot block devices can be organized with either 8-bit or 16-bit bus.



Block diagram of the 28F004/28F400



9.5 FLASH Memory

■ Standard boot block FLASH memories

- ❖ Another new feature introduced with the boot block architecture is that of a hardware-lockable block. In the 28F004/28F400, the 16Kbyte boot block can be locked by applying logic 0 to the write protected input (WP).

7FFFFH	16K-byte Boot block	7FFFFH	128K-byte Main block
7C000H		60000H	
7BFFFFH	8K-byte Parameter block	5FFFFH	128K-byte Main block
7A000H		40000H	
79FFFFH	8K-byte Parameter block	3FFFFH	128K-byte Main block
78000H		20000H	
77FFFFH	96K-byte Main block	1FFFFH	96K-byte Main block
60000H		08000H	
5FFFFH	128K-byte Main block	07FFFFH	8K-byte Parameter block
40000H		06000H	8K-byte Parameter block
3FFFFH	128K-byte Main block	05FFFFH	8K-byte Parameter block
20000H		04000H	16K-byte Boot block
1FFFFH	128K-byte Main block	03FFFFH	
00000H		00000H	

28F004-T

28F004-B

Note:
Address = A[18:0]

Top and bottom boot block organization of the 28F004





9.5 FLASH Memory

■ Standard boot block FLASH memories

- ❖ If the 28F400 device is not in use, it can be put into the deep power-down mode to conserve power by switch RP (Reset/Deep power-down) input to logic 0.
- ❖ The 28F004/28F400 uses a command user interface (CUI), status register, and write-state machine to initiate an internally implemented and highly automated method of erasing and programming the blocks of the storage array. This is known as **automatic erase and write**.



9.5 FLASH Memory

■ Standard boot block FLASH memories

Command	Notes	First Bus Cycle			Second Bus Cycle		
		Oper	Addr	Data	Oper	Addr	Data
Read array	8	Write	X	FFH			
Intelligent identifier	1	Write	X	90H	Read	IA	IID
Read status register	2,4	Write	X	70H	Read	X	SRD
Clear status register	3	Write	X	50H			
Word/byte write		Write	WA	40H	Write	WA	WD
Alternate word/byte write	6,7	Write	WA	10H	Write	WA	WD
Block erase/confirm	6,7	Write	BA	20H	Write	BA	DOH
Erase suspend/resume	5	Write	X	B0H	Write	X	DOH

Address

BA = Block Address
IA = Identifier Address
WA = Write Address
X = Don't Care

Data

SRD = Status Register Data
IID = Identifier Data
WD = Write Data

Notes:

- Bus operations are defined in Tables 4 and 5.
- IA = Identifier Address: A0 = 0 for manufacturer code, A0 = 1 for device code.
- SRD—Data read from Status Register.
- IID = Intelligent Identifier Data. Following the Intelligent Identifier command, two Read operations access manufacturer and device codes.
- BA = Address within the block being erased.
- WA = Address to be written. WD = Data to be written at location WD.
- Either 40H or 10H commands is valid.
- When writing commands to the device, the upper data bus [DQ₈–DQ₁₅] = X (28F400 only) which is either V_{IL} or V_{IH}, to minimize current draw.

28F004 command bus definition



9.5 FLASH Memory

Standard boot block FLASH memories

WSMS	ESS	ES	DWS	VPPS	R	R	R
7	6	5	4	3	2	1	0
Notes: SR.7 = WRITE STATE MACHINE STATUS (WSMS) 1 = Ready 0 = Busy SR.6 = ERASE-SUSPEND STATUS (ESS) 1 = Erase Suspend 0 = Erase in Progress/Completed SR.5 = ERASE STATUS 1 = Error in Block Erasure 0 = Successful Block Erase SR.4 = PROGRAM STATUS 1 = Error in Byte/Word Program 0 = Successful Byte/Word Program SR.3 = V_{PP} STATUS 1 = V_{PP} Low Detect, Operation Abort 0 = V_{PP} OK SR.2-SR.0 = RESERVED FOR FUTURE ENHANCEMENTS							
				Write State Machine bit must first be checked to determine Byte/Word program or Block Erase completion before the Program or Erase Status bits are checked for success. When Erase Suspend is issued, WSM halts execution and sets both WSMS and ESS bits to "1." ESS bit remains set to "1" until an Erase Resume command is issued. When this bit is set to "1," WSM has applied the maximum number of erase pulses to the block and is still unable to successfully verify block erasure. When this bit is set to "1," WSM has attempted but failed to program a byte or word. The V_{PP} Status bit, unlike an A/D converter, does not provide continuous indication of V_{PP} level. The WSM interrogates V_{PP} level only after the Byte Write or Erase command sequences have been entered, and informs the system if V_{PP} has not been switched on. The V_{PP} Status bit is not guaranteed to report accurate feedback between V_{PP(L)} and V_{PP(H)}. These bits are reserved for future use and should be masked out when polling the Status Register.			

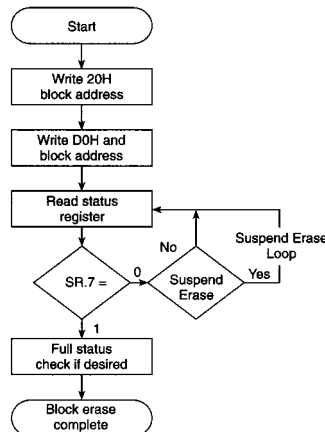
Status register bit definition

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9.5 FLASH Memory

Standard boot block FLASH memories



Bus Operation	Command	Comments
Write	Erase setup	Data = 20H Addr = Within block to be erased
Write	Erase confirm	Data = D0H Addr = Within block to be erased
Read		Status register data toggle CE# or OE# to update status register
Standby		Check SR.7 1 = WSM ready 0 = WSM busy
Repeat for subsequent block erasures. Full Status Check can be done after each block erase, or after a sequence of block erasures. Write FFH after the last operation to reset device to read array mode.		

Erase operation flowchart and bus activity

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9.5 FLASH Memory

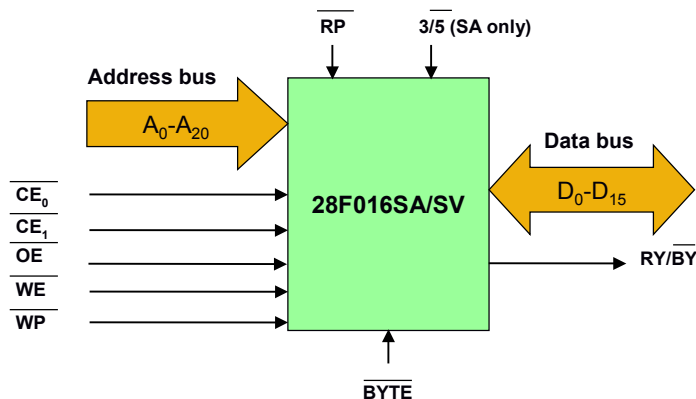
■ Standard FlashFile FLASH memories

- ❖ The highest-density FLASH memories available today are those designed with the FlashFile architecture.
- ❖ FlashFile memories are intended for use in large-code storage applications and to implement solid-state mass-storage devices such as the FLASH card and FLASH drive.
- ❖ The FlashFile memories support block locking. The blocks are independently programmable as locked or unlocked.



9.5 FLASH Memory

■ Standard FlashFile FLASH memories

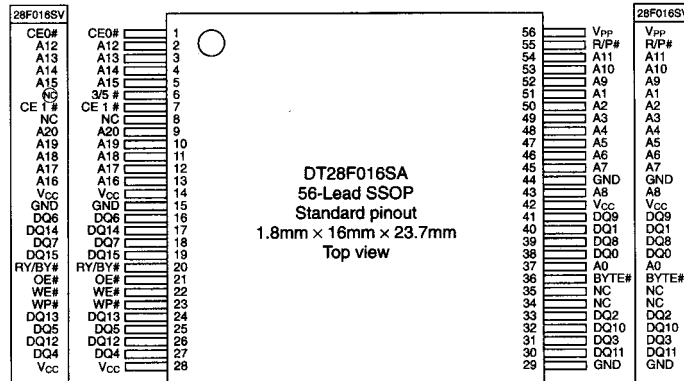


Block diagram of the 28F016SA/SV



9.5 FLASH Memory

■ Standard FlashFile FLASH memories



Pin lay-out of the SSOP 28F016SA/SV



9.5 FLASH Memory

■ Standard FlashFile FLASH memories

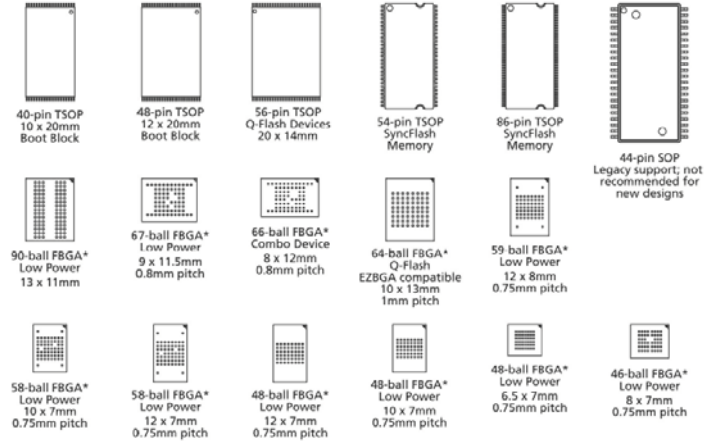
1FFFFF	64K-byte block	31
1F5000	64K-byte block	30
1E0000	64K-byte block	29
1D5000	64K-byte block	28
1C0000	64K-byte block	27
1B5000	64K-byte block	26
1A0000	64K-byte block	25
195000	64K-byte block	24
180000	64K-byte block	23
175000	64K-byte block	22
160000	64K-byte block	21
155000	64K-byte block	20
140000	64K-byte block	19
135000	64K-byte block	18
120000	64K-byte block	17
115000	64K-byte block	16
100000	64K-byte block	15
FF5000	64K-byte block	14
FF0000	64K-byte block	13
FC5000	64K-byte block	12
FC0000	64K-byte block	11
F85000	64K-byte block	10
F80000	64K-byte block	9
F75000	64K-byte block	8
F70000	64K-byte block	7
F65000	64K-byte block	6
F60000	64K-byte block	5
F55000	64K-byte block	4
F50000	64K-byte block	3
F45000	64K-byte block	2
F40000	64K-byte block	1
F35000	64K-byte block	0
000000		

Byte-wide mode memory
map of the 28F016SA/SV



9.5 FLASH Memory

■ FLASH packages



Source: Micron Technology, Inc.,

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9.5 FLASH Memory

■ FLASH memory applications

- ❖ Digital cellular phones
- ❖ PDAs
- ❖ Digital cameras
- ❖ LAN switches
- ❖ Digital set-top boxes
- ❖ Embedded controllers
- ❖ BIOS
- ❖ FLASH disk



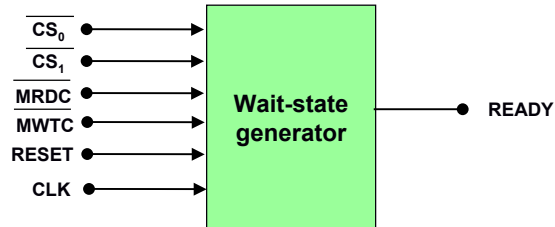
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9.6 Wait-State Circuitry

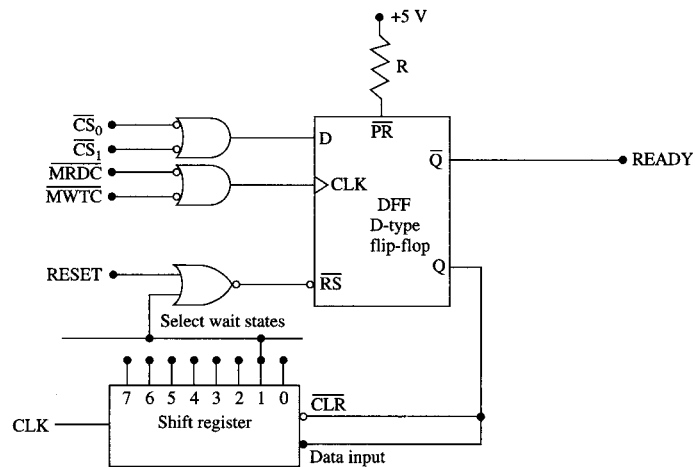
- Depending on the access time of the memory devices used and the clock rate of the MPU, a number of wait states may need to be inserted into external memory read and write operations.



Wait-state generator circuit block diagram



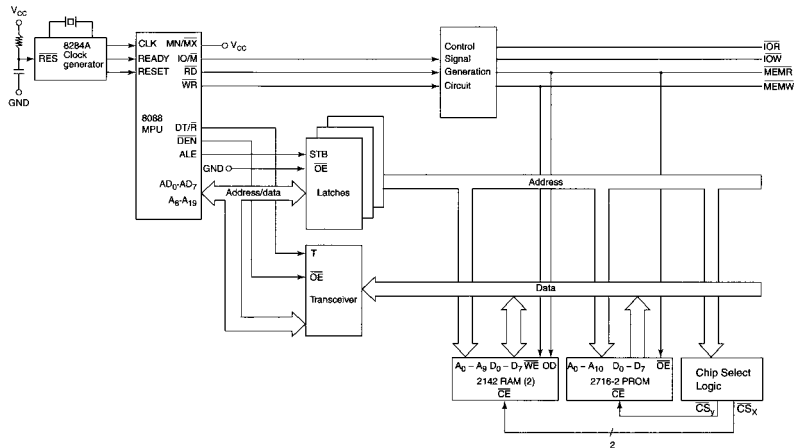
9.6 Wait-State Circuitry



Typical wait-state generator circuit



9.7 8088/8086 Microcomputer System Memory Circuitry

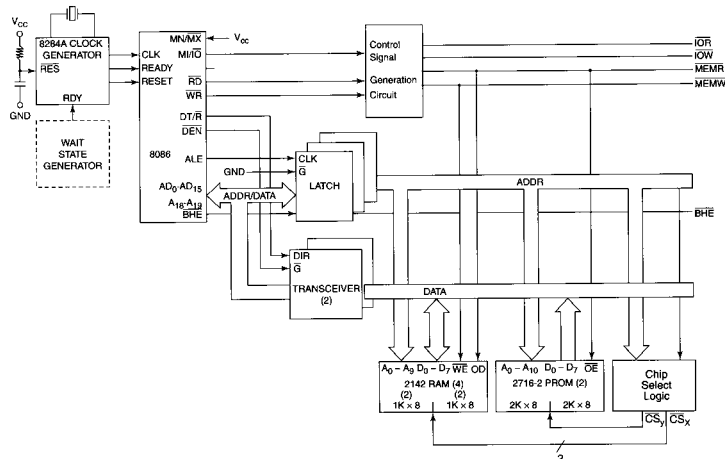


Minimum-mode 8088 system memory interface

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9.7 8088/8086 Microcomputer System Memory Circuitry

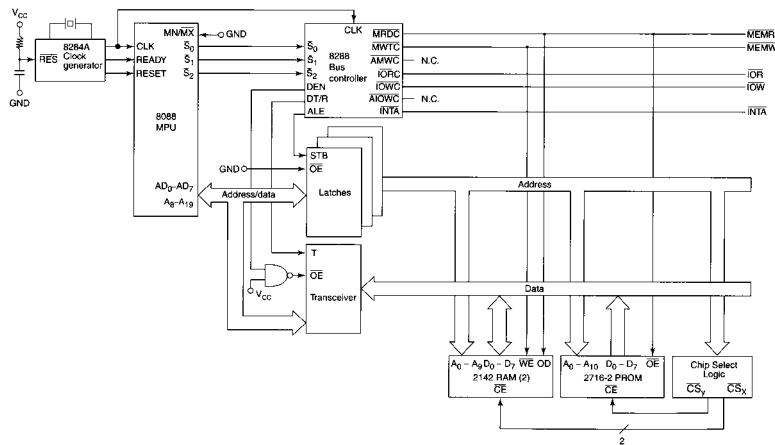


Minimum-mode 8086 system memory interface

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9.7 8088/8086 Microcomputer System Memory Circuitry



Maximum-mode 8088 system memory interface

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9.7 8088/8086 Microcomputer System Memory Circuitry

■ Program storage memory

- ❖ Attaching several EPROM devices to the system bus expands the capacity of program storage memory.
- ❖ High-order bits of the 8088's address are decoded to produce chip-select signals. Each chip-select is applied to the $\overline{CE}$ (chip-enable) input of the EPROM.
- ❖ In the maximum-mode circuit, the 8288 bus controller, rather than the 8088, produces the control signals for the address latches and data bus transceiver.

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9.7 8088/8086 Microcomputer System Memory Circuitry

■ Data storage memory

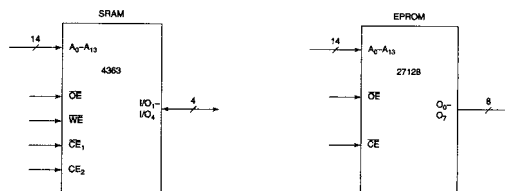
- ❖ Information that frequently changes is normally implemented with random access read/write memory (RAM).
- ❖ If the amount of memory required in the microcomputer is small, the memory subsystem is usually designed with SRAMs.
- ❖ DRAMs require refresh support circuit which is not warranted if storage requirement are small.



9.7 8088/8086 Microcomputer System Memory Circuitry

EXAMPLE

Design a memory system consisting of 32Kbytes of R/W memory and 32Kbytes of ROM memory. Use SRAM devices to implement R/W memory and EPROM devices to implement ROM memory. The memory devices to be used are shown below. R/W memory is to reside over the address range 00000_{16} through $07FFF_{16}$ and the address range of ROM memory is to be $F8000_{16}$ through $FFFFF_{16}$. Assume that the 8088 microprocessor system bus signals that follow are available for use: A_0 through A_{19} , D_0 through D_7 , $\overline{\text{MEMR}}$, $\overline{\text{MEMW}}$.



9.7 8088/8086 Microcomputer System Memory Circuitry

SOLUTION:

First let us determine the number of SRAM devices needed.

$$\text{No. of SRAM devices} = 32\text{Kbyte} / (16\text{K} \times 4) = 4$$

To provide an 8-bit data bus, two SRAMs must be connected in parallel. Two pairs connected in this way are then placed in series to implement the R/W address range, and each pair implements 16Kbytes.

Next let us determine the number of EPROM devices needed.

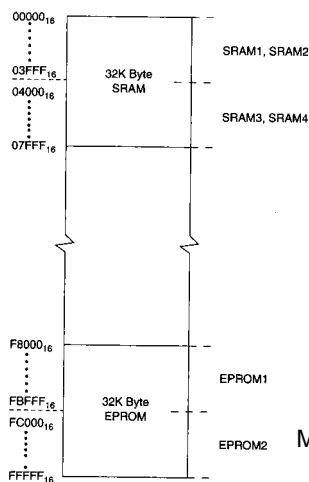
$$\text{No. of EPROM devices} = 32\text{Kbyte} / 16\text{Kbyte} = 2$$

These two devices must be connected in series to implement the ROM address range and each implement 16Kbytes of storage.



9.7 8088/8086 Microcomputer System Memory Circuitry

SOLUTION:

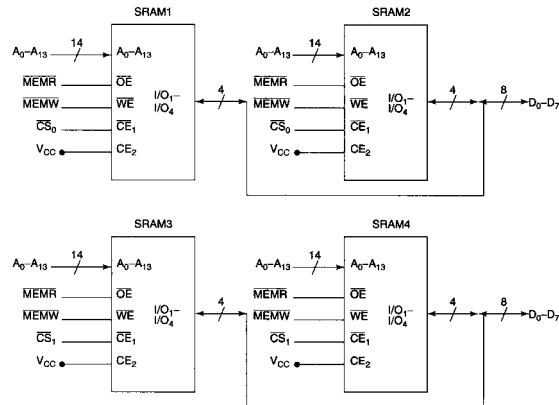


Memory map of the system



9.7 8088/8086 Microcomputer System Memory Circuitry

SOLUTION:

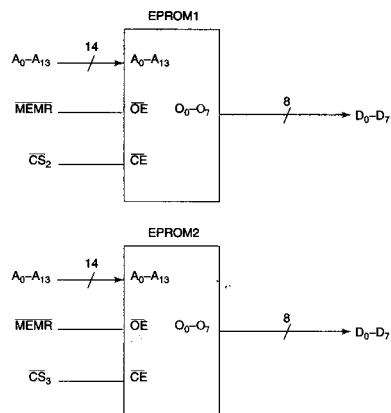


RAM memory organization for the system design



9.7 8088/8086 Microcomputer System Memory Circuitry

SOLUTION:



ROM memory organization for the system design



9.7 8088/8086 Microcomputer System Memory Circuitry

SOLUTION:

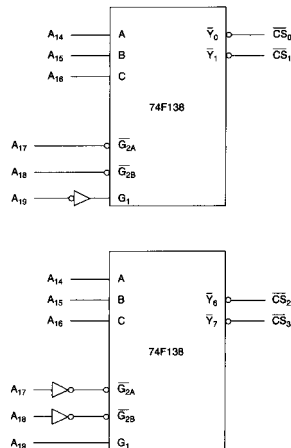
$$\begin{aligned}
 &A_{19} \dots A_0 \\
 00000_{16} &= 0000 \ 0000 \ 0000 \ 0000 \ 0000 \\
 &\vdots \\
 03FFF_{16} &= \underbrace{0000 \ 0011}_{\overline{CS}_0} \ 1111 \ 1111 \ 1111 \\
 &\vdots \\
 04000_{16} &= 0000 \ 0100 \ 0000 \ 0000 \ 0000 \\
 &\vdots \\
 07FFF_{16} &= \underbrace{0000 \ 0111}_{\overline{CS}_1} \ 1111 \ 1111 \ 1111 \\
 &\vdots \\
 F8000_{16} &= 1111 \ 1000 \ 0000 \ 0000 \ 0000 \\
 &\vdots \\
 FFFFF_{16} &= \underbrace{1111 \ 1011}_{\overline{CS}_2} \ 1111 \ 1111 \ 1111 \\
 &\vdots \\
 FC000_{16} &= 1111 \ 1100 \ 0000 \ 0000 \ 0000 \\
 &\vdots \\
 FFFFF_{16} &= \underbrace{1111 \ 1111}_{\overline{CS}_3} \ 1111 \ 1111 \ 1111
 \end{aligned}$$

Address range analysis for the design of chip select signals



9.7 8088/8086 Microcomputer System Memory Circuitry

SOLUTION:



Chip-select logic

